

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Chemistry — Class 9 (Science Group)

Syllabus: Chapter 1 — Nature of Chemistry | Chapter 2 — Matter | Chapter 3 — Atomic Structure | Chapter 4 — Periodic Table & Periodicity | Chapter 5 — Chemical Bonding | Chapter 6 — Stoichiometry | Chapter 7 — Electrochemistry | Chapter 8 — Energetics | Chapter 9 — Chemical Equilibrium

Total Marks: 43

Time: 2 Hours

Paper: Exam-19 (Version 1)

SECTION A — MCQ Answer Key (24 × 1 = 24 Marks)

#	Question	Correct Answer	Explanation	Source in her textbook
1	Chemistry is best defined as the study of	B. matter, its properties, composition and its interactions with other matter and energy 1	This is the definition given in the Chapter 1 learning outcomes. Option A describes biology, C describes physics and D describes geology.	Ch1 SLO 1 (definition of chemistry)
2	State of matter compressed most easily	C. gas 1	Gases have very large spaces between their particles, so they are highly compressible. Solids and liquids have closely packed particles and are almost incompressible.	Ch2 §2.1, macroscopic properties (density, compressibility, fluidity)
3	Which one is a compound?	C. carbon dioxide 1	Carbon dioxide is made of two different elements chemically combined in a fixed ratio. Air, brass and sea water are all mixtures, whose components are not chemically combined.	Ch2 SLO (elements, compounds and mixtures)
4	Solution that can dissolve no more solute	A. saturated 1	A saturated solution contains the maximum amount of solute that can dissolve at that temperature. An unsaturated solution can still dissolve more.	Ch2 §2.5, saturated and unsaturated solutions
5	Force holding protons and neutrons in the nucleus	D. strong nuclear force 1	The book states that the nucleus is made up of protons and neutrons held together by the strong nuclear force. It is strong enough to overcome the repulsion between the positively charged protons.	Ch3 SLO (nucleus held together by strong nuclear force)
6	Effect of increasing shell number	B. both increase 1	The learning outcome states that orbits (shells) are energy levels of electrons, and a larger shell implies higher energy and a greater average distance from the nucleus.	Ch3 SLO (shells as energy levels)

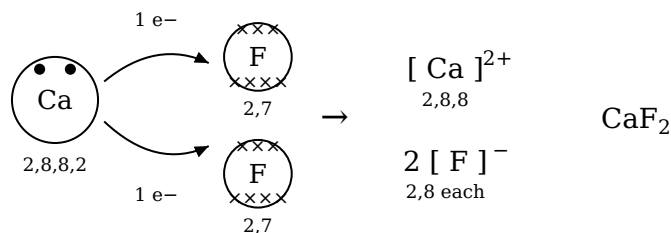
7	Nucleon number (mass number) is the sum of	A. protons and neutrons 1	The nucleon number is defined as the sum of the protons and neutrons in the nucleus. Electrons are not counted because their mass is negligible.	<i>Ch3 SLO (nucleon number / atomic mass)</i>
8	Two isotopes of the same element have	C. the same chemical properties but different molecular masses 1	Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons. The book states isotopes can affect molecular mass but not chemical properties, since chemistry depends on the electrons.	<i>Ch3 SLO (isotopes affect molecular mass but not chemical properties)</i>
9	Modern periodic table order of increasing	B. proton number 1	The periodic table is defined as an arrangement of elements in periods and groups in order of increasing proton number (atomic number), not atomic mass.	<i>Ch4 SLO 1 (definition of the periodic table)</i>
10	Charge on the ion formed by a Group 2 element	D. +2 1	A Group 2 element has 2 electrons in its outermost shell. It loses both to reach a noble gas configuration, so it forms an ion of charge +2.	<i>Ch4 SLO (group number and the charge of ions formed)</i>
11	Atomic radius on moving down a group	A. increases 1	Going down a group a new shell is added at each step, so the outermost electrons lie further from the nucleus and the atomic radius increases.	<i>Ch4 SLO (trends in groups and periods, atomic radius)</i>
12	Collective name for Group 17	C. halogens 1	Group 17 elements (F, Cl, Br, I) are called the halogens. Group 1 are alkali metals, Group 2 alkaline earth metals and Group 18 noble gases.	<i>Ch4 SLO (alkali metals, halogens, noble gases ...)</i>
13	A metallic bond is the attraction between	B. a sea of delocalized electrons and positively charged metal cations 1	This is the definition in the Chapter 5 learning outcomes. Option A defines an ionic bond and option C defines a covalent bond.	<i>Ch5 SLO (definition of the metallic bond)</i>
14	Metals are electropositive because they tend to	D. lose electrons 1	Metals have few electrons in their outermost shell and low ionization energies, so they readily lose electrons to form positive ions. That is what electropositive means.	<i>Ch5 SLO (electropositive and electronegative nature of metals)</i>
15	Ionic compared with covalent compounds	A. higher melting points 1	Ionic compounds have strong electrostatic forces between oppositely charged ions throughout a giant lattice, so much more heat is needed to separate them than for the weak forces	<i>Ch5 SLO (differentiate ionic and covalent compounds)</i>

			between covalent molecules.	
16	Atoms bond to achieve the configuration of the nearest	C. noble gas 1	Atoms gain, lose or share electrons to obtain the stable noble gas electronic configuration, which is the basis of the octet and duplet rules.	Ch5 SLO (noble gas configuration, octet and duplet rules)
17	The empirical formula gives	D. the simplest whole number ratio of the different atoms 1	This is the textbook definition. Option A is the definition of the molecular formula, which gives the actual number and type of atoms in one molecule.	Ch6 SLO (definition of empirical formula)
18	Moles in 80 g of NaOH	B. 2 mol 1	Molar mass of NaOH = 23 + 16 + 1 = 40 g/mol. Number of moles = mass ÷ molar mass = 80 ÷ 40 = 2 mol.	Ch6 SLO (amount of substance = mass / molar mass)
19	Meaning of the state symbol (aq)	A. dissolved in water 1	(aq) stands for aqueous, meaning the substance is dissolved in water. The other symbols are (s) solid, (l) pure liquid and (g) gas.	Ch6 SLO (construct chemical equations including state symbols)
20	Formula from Ca^{2+} and NO_3^-	D. $\text{Ca}(\text{NO}_3)_2$ 1	The compound must be electrically neutral. One Ca^{2+} (+2) needs two NO_3^- ($2 \times -1 = -2$). Brackets are required around the polyatomic ion before the subscript.	Ch6 SLO (deduce the formula of a binary compound from ions)
21	Balancing $\text{Zn} + \text{O}_2 \rightarrow \text{ZnO}$	B. 2, 1, 2 1	$2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$. Left: 2 Zn and 2 O. Right: 2 Zn and 2 O. Only option B balances both elements.	Ch6 SLO (construct and balance chemical equations)
22	Oxidation number of an element in its free state	C. zero 1	An uncombined element has not gained, lost or shared electrons with any other element, so its oxidation number is zero. Examples: Zn, O_2 , Cl_2 , S_8 .	Ch7 SLO (oxidation number of elements in their free state is zero)
23	Oxidation number of a monoatomic ion	A. the same as the charge on the ion 1	A monoatomic ion is formed by the loss or gain of electrons by one atom, so its oxidation number is exactly its ionic charge. For example Na^+ is +1 and Cl^- is -1.	Ch7 SLO (oxidation number of a monoatomic ion = charge on the ion)
24	Activation energy is the minimum energy for a	D. successful collision 1	The learning outcome defines activation energy as the minimum energy that colliding molecules must have for a successful collision. Collisions with less energy than this do not lead to a reaction.	Ch8 SLO (definition of activation energy)

SECTION B – Short Questions (3 × 3 = 9 Marks)

Q.2 (i) Formation of calcium fluoride, CaF₂ 3 marks**(a) Dot-and-cross diagram** 2

Calcium (Z = 20) has the configuration 2, 8, 8, 2 and so has 2 electrons in its outermost shell. Fluorine (Z = 9) has the configuration 2, 7 and needs 1 more electron. One calcium atom therefore transfers one electron to each of **two** fluorine atoms.



Calcium loses 2 electrons (dots) and each fluorine gains 1 (crosses), giving Ca²⁺ and 2F⁻.

Award **1** for showing Ca losing 2 electrons to become Ca²⁺ with the configuration 2,8,8, and **1** for showing **two** fluorine atoms each gaining 1 electron to become F⁻ with the configuration 2,8.

(b) Type of bond 1

An **ionic bond**, which is the strong electrostatic force of attraction between oppositely charged ions (here Ca²⁺ and F⁻). **1**

Source: Ch5 SLOs (compare the formation of cations and anions; define the ionic bond as a strong electrostatic attraction between oppositely charged ions). Accept dots and crosses drawn either way round, and accept the configurations written as 2,8,8,2 or 1s²2s²2p⁶3s²3p⁶4s². Do not deduct if only the outermost shells are drawn.

Q.2 (ii) Relative atomic mass of lithium 3 marks**(a) Relative atomic mass** 2

Relative atomic mass = (mass of isotope 1 × % abundance) + (mass of isotope 2 × % abundance), all divided by 100. **1**

$$\text{R.A.M.} = [(6 \times 7.5) + (7 \times 92.5)] \div 100$$

$$\text{R.A.M.} = (45 + 647.5) \div 100 = 692.5 \div 100$$

$$\text{R.A.M. of lithium} = 6.925 \approx 6.9 \quad \mathbf{1}$$

(b) Particles in one atom of ⁷Li 1

Lithium has atomic number 3, so protons = **3**, electrons = **3**, and neutrons = mass number – atomic number = 7 – 3 = **4**. **1**

Source: Ch3 SLOs (define relative atomic mass as the average mass of the isotopes of an element compared to 1/12th of the mass of carbon-12; calculate relative atomic mass from relative masses and abundance of isotopes; determine the number of protons and neutrons of different isotopes). Accept 6.9 or 6.93; award the formula mark even if the arithmetic slips.

Q.2 (iii) Beryllium, Z = 4 3 marks**(a) Electronic configuration** 1**1s² 2s²** (shell notation 2, 2). 1**(b) Group and period** 1**Group 2**, because it has 2 electrons in its outermost shell. **Period 2**, because its electrons occupy 2 shells. 1**(c) Charge on its ion** 1

Beryllium forms an ion of charge **+2** (Be²⁺). Its group number is 2, which tells us it has 2 electrons in its outermost shell; it loses both of them to reach the stable configuration of the nearest noble gas (helium, 1s²), leaving a 2+ charge. 1

Source: Ch4 SLOs (identify the group or period of an element using its electronic configuration; explain the relationship between group number and the charge of ions formed from elements in the group in terms of their outermost shells). Accept "2,2" for (a).

SECTION C — Long Questions (2 × 5 = 10 Marks)**Q.3 (i) Empirical and molecular formula from percentage composition** 5 marks**(a) Definitions** 1

Empirical formula: the formula that gives the simplest whole number ratio of the different atoms in a compound.

Molecular formula: the formula that gives the actual number and type of the different atoms present in one molecule of the compound. 1

(b) Determining the formulae 3**Step 1 — divide each percentage by the relative atomic mass:** 1

$$\text{K: } 56.5 \div 39 = 1.449$$

$$\text{C: } 8.7 \div 12 = 0.725$$

$$\text{O: } 34.8 \div 16 = 2.175$$

Step 2 — divide each answer by the smallest (0.725): 1

$$\text{K: } 1.449 \div 0.725 = 2$$

$$\text{C: } 0.725 \div 0.725 = 1$$

$$\text{O: } 2.175 \div 0.725 = 3$$

Simplest whole number ratio K : C : O = 2 : 1 : 3

Empirical formula = K₂CO₃

Step 3 — compare with the relative formula mass: 1

$$\text{Empirical formula mass} = 2(39) + 12 + 3(16) = 78 + 12 + 48 = 138$$

$$n = \text{relative formula mass} \div \text{empirical formula mass} = 138 \div 138 = 1$$

Molecular formula = K₂CO₃ (potassium carbonate)

(c) Balanced equation with state symbols 1

Both the balancing (1, 2, 2, 1, 1) and all five state symbols are needed for the mark. Accept K₂CO₃(s) if the candidate treats the white solid as undissolved.

Source: Ch6 SLOs (define empirical and molecular formula; use amount of substance = mass / molar mass; construct chemical equations to show reactants forming products, including state symbols). The percentages were chosen so that Step 2 divides exactly; no rounding judgement is required.

Q.3 (ii) Mole calculation for magnesium nitrate 5 marks

(a) Molar mass of $\text{Mg}(\text{NO}_3)_2$ 1

$$M = 24 + 2 \times [14 + (3 \times 16)] = 24 + 2 \times (14 + 48) = 24 + (2 \times 62) = 24 + 124$$

$$\text{Molar mass} = 148 \text{ g/mol}$$
 1

(b) Number of moles in 22.2 g 2

$$\text{Number of moles} = \text{mass} \div \text{molar mass}$$
 1

$$n = 22.2 \text{ g} \div 148 \text{ g/mol}$$

$$n = 0.15 \text{ mol}$$
 1

(c) Number of formula units 1

$$\text{Number of formula units} = \text{number of moles} \times N_A$$

$$= 0.15 \times 6.022 \times 10^{23}$$

$$= 9.033 \times 10^{22} \approx 9.03 \times 10^{22} \text{ formula units}$$
 1

(d) Oxidation number of magnesium 1

The oxidation number of magnesium is **+2**. The rule is that **the oxidation number of a monoatomic ion is the same as the charge on that ion**: magnesium is in Group 2, so it loses 2 electrons and exists here as Mg^{2+} . 1

(A candidate may instead argue from the rule that the oxidation numbers in a neutral compound sum to zero: 2 nitrate ions carry $2 \times -1 = -2$, so Mg must be +2. Accept either.)

Source: Ch6 SLOs (use the relationship $\text{amount of substance} = \text{mass} / \text{molar mass}$ to calculate number of moles, mass and number of particles; define the mole as the amount of substance containing Avogadro's number of particles) and Ch7 SLO (identify that the oxidation number of a monoatomic ion is the same as the charge on the ion). Accept 9.0×10^{22} or 9.03×10^{22} .

— End of Answer Key —